MIL-T-14757 (MU)
10 May 1968

MILITARY SPECIFICATION

TRICHLOROTRIFLUOROETHANE ($C Cl_2 FC Cl F_2$)
(FOR USE IN AMMUNITION)

1. SCOPE

1.1 This specification covers one type of chlorofluoro hydrocarbon for use in Aerial Mines (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials - ASTM Standards

FSC: 1345

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(Copies may be purchased directly from ASTM, 1916 Race Street, Philadelphia 3, Pa.)

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for Transportation of Explosives and Other Dangerous Articles

(Note: In particular, attention shall be paid to ICC 17E, United P, 18-55-56.)

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition--revised 1956) available from the Superintendant of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publications should cite "49 CFR 71-90 (Rev. 1956)".)

3. REQUIREMENTS

3.1 Material.-The material shall be a trichlorotrifluoro hydrocarbon with the following properties:

Appearance:	Clear, colorless liquid (see 4.3.1)
Boiling point:	47.6°C ± 0.5°C (Standard Barometric Pressure (see 4.3.3))
Purity:	99.8 Percent, minimum (see 4.3.7)
Residue:	5 ppm by weight, maximum (see 4.3.4)
Acid Number:	0.003 maximum (see 4.3.6)
Moisture:	30 ppm by weight, maximum (see 4.3.5 & 6.5)
Copper:	0.1 ppm by weight, maximum (see 4.3.8)

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein.

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4.1.1 Contractor quality assurance system.-If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, (measuring and testing equipment,) and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedures of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure which affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all requirements and provisions of the approved current written description of the material.

b. Specification number and date, together with an identification and date of changes.

c. Certificates of analysis on all material used directly by the contractor when such material is controlled by Government specification shall be made available upon request by the contracting officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Product inspection.-Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provisions of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractor's records are reliable.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of one or more batches of the material produced by one manufacturer in accordance with the same specification, or same specification revision under one continuous set of operating conditions. Each batch shall consist of that quantity of the material that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235 when applicable).

4.2.2 Examination.-Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans, in accordance with Standard MIL-STD-1235, may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect.

4.2.2.1 Drums

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Leaking drum	Visual	01001
102.	Improper type of drum (see Section 5)	Visual	01002
Minor:	AQL 0.65 Percent		
201.	Marking missing, incorrect or illegible	Visual	01003

4.2.3 Sampling

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4.2.3.1 Material in drums.-Except as noted below (see 4.2.3.2) a quantity of at least two (2) gallons shall be taken from the lot. This shall be accomplished by taking equal quantities from each drum of the sample which shall be in accordance with Table I:

TABLE I

Quantity in lot	Number sampled
Less than 50 drums	10
51 to 100	15
Over 101	20

The portions taken shall be mixed and tested for conformance to 3.1.

4.2.3.2 Additional Tests.-In addition whenever delivery is made in drums rather than tank car or tank truck, individual 200 ml samples shall be taken from ten randomly selected shipping containers (in addition to the sample specified in 4.2.3.1) and each sample shall be tested individually for boiling range or boiling point as specified in 4.3.2 or 4.3.3, as applicable. If any of these samples fails to comply with the applicable requirement of 3.1, the lot shall be rejected.

4.2.3.3 Materials shipped in tank cars.-A representative sample shall be withdrawn from the tank car using a suitable thief. The total quantity of the sample shall be at least 5 gallons. Test this sample in accordance with the methods specified in 4.3. If this sample fails to comply with any of the requirements specified in 3.1, the lot shall be rejected.

4.3 Test methods and procedures

4.3.1 Appearance - Major Defect - (Defect Code No. 02001).-The sample shall be visually inspected for presence of solids and color.

4.3.2 Boiling range - Major Defect - (Defect Code No. 03001).- Distill 100 ml of the sample in a 500 ml round-bottom distilling flask equipped with a thermometer well and a water cooled condensor connected to a 100 ml graduate. Use a warm water bath to provide a gentle boiling. Add a few boiling chips to the flask to promote smooth boiling. The temperature should be constantly observed after the first 5 ml of sample has been distilled until 85 ml of the sample has been distilled. (Note: 1. Equivalent equipment may be used, and 2. do not use stop cock grease.)

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4.3.3 Boiling point - Major Defect - (Defect Code No. 04001).-Reflux 100 ml of the sample in a 500 ml round bottom reflux flask equipped with a thermometer well and a water cooled condensor. Adjust the temperature of the electric heating mantle to provide gentle refluxing. Add a few boiling chips to promote smooth boiling. Record the temperature after refluxing for thirty minutes.
(Note: 1. Equivalent equipment may be used, and 2. do not use stop cock grease.)

4.3.4 Residue - Major Defect - (Defect Code No. 05001).

4.3.4.1 Apparatus (Note: Equivalent apparatus may be used.)

Gram electrobalance, Bahn, Fisher Scientific Co.
Graduated Cylinder - 1 liter capacity
Beaker 250-ml capacity, tall form
Wash bottle, Guth type with Swinny adapter
Aluminum dishes
Hot plate
Filters, 1.20 micron pore size, 13mm diameter

4.3.4.2 Procedure.-Tare and weigh to 0.01 mg a dry aluminum weighing dish. Measure 1000 ml of the sample into a clean 1-liter graduated cylinder. Fill a carefully cleaned 250 ml beaker about 3/4 full with sample from the graduate. Cover the beaker with a watch glass suspended on a glass rod support. Place the beaker on a hot plate and boil gently until the volume is reduced to approximately 20 ml. Refill the beaker and repeat the above procedure until all the sample has been evaporated. Wash quantitatively the residue from the beaker into the tared aluminum dish using "Freon" Precision Cleaning Agent. Use a Guth-type unitized wash bottle fitted with a Swinny adapter and a 1.20 micron Millipore filter as the container for the washing solvent. Evaporate the washings in the aluminum dish on a hot plate adjusted to low temperature (since several washings will be necessary for complete removal of the residue, evaporate the Freon after each washing). Place the aluminum dish in an oven at 100 plus or minus 5°C for 1/2 hour after the final evaporation to dryness. Cool the dish and reweigh to 0.01 mg. Calculate the ppm of total residue as follows:

$$\text{ppm total residue} = \frac{(A-B)\ 10^3}{W}$$

where:

A = final weight of dish and residue, mg
B = initial weight of dish, mg
W = weight of sample, g

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4.3.5 Moisture - Major Defect - (Defect Code No. 06001).-Determine the moisture content of a 100 gram sample of the material using an appropriate Karl Fischer Method. Calculate the results in ppm. A recommended method is the method described in ASTM-D 1123.

4.3.6 Acid Number - Major Defect - (Defect Code No. 07001).

4.3.6.1 Apparatus

10 ml Buret graduated in 0.02 subdivisions w/Teflon stop cock and syringe needle tip adapter

4.3.6.2 Reagents

Isopropyl alcohol, anhydrous
Potassium hydroxide, in isopropanol, 0.01N standard solution
Phenolphthalein indicator (0.05 g/100 ml isopropyl alcohol)

4.3.6.3 Procedure

Add 20 ml of isopropyl alcohol in a 250 ml Erlenmeyer flask, add 3 drops of phenolphthalein indicator solution and titrate to a pink end point with the alcoholic potassium hydroxide. The end point should last for at least 15 seconds. Stopper the flask. Pour a weighed 50 ml sample of the material into the flask. Titrate the sample with the standard potassium hydroxide solution to a 15 second pink end point. Record the amount of titrant required.

$$\text{Total acid number, mg of KOH/g sample} = \frac{A \times N \times 56.1}{W}$$

where:

A = ml of potassium hydroxide titrant
N = normality of KOH
W = weight of sample, gms

4.3.7 Purity - Major Defect - (Defect Code No. 08001).

4.3.7.1 Apparatus

A gas chromatograph of adequate sensitivity which can be temperature programmed from 30 to 150°C. A Burrel Kromotag K1 with manual programming has been found satisfactory.
Flow meter, soap-bubble type
Chromatograph column: 2 1/2 meters x 1/4 inch diameter, packed with Perkin-Elmer Type R
Hypodermic syringe, 0.1 ml.

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4.3.7.2 Reagents

Helium, research grade
Column packing: 20% by weight of polypropylene gycol (avg. mol. wt. 550) on 60-80 mesh Chromosorb solid support

4.3.7.3 Adjust the helium flow to approximately 100 ml/min and maintain the selected flow rate to within 1%; condition the column at 160°C until a stable base line is obtained. Cool the column to 30°C. Fill a 0.1 ml hypodermic syringe with 0.050 ml of sample. Introduce sample into sample port and note the time. After 630 seconds, start temperature programming the column at rate of 15°C/min to a maximum temperature of 150°C. Once the correct time is established for a column and instrument, temperature programming should be started. Allow the sample to elute until all possible impurities have been removed from the column (approximately 20 min.) Use the minimum attenuation which will keep recorder on scale at all times. The calculations shall be made as follows after measuring the area of each peak:

$$\text{Purity, Wt. \%} = \frac{\text{Peak area of material} \times 100}{\text{Peak area of material and peak areas of all impurities}}$$

where:

Peak area = height x width at 0.5 peak x attenuator setting

4.3.8 Copper - Major Defect - (Defect Code No. 09001).-Evaporate 1000 ml of the sample in a clean suitable laboratory vessel. Test the residue for copper using a suitable method (capable of detecting .1 ppm copper) appearing in a standard textbook or standard. Prior to use of this method approval shall be obtained from Picatinny Arsenal, Dover, New Jersey ATTN: SMUPA-ND8, thru the Contracting Officer.

5. PREPARATION FOR DELIVERY

(Note: The packing described in 5.1.1 is advisory rather than mandatory. If the supplier wishes to deviate from the provisions in 5.1.1 he shall obtain the approval of the procuring activity.)

5.1 Packing

5.1.1 Drums.-Unless otherwise specified in the contract or purchase order, the material shall be packed in non-returnable drums in accordance with ICC-17E having a sterilkote, 46 series high baled phenolic modified epoxy lining.

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5.1.2 Level C.-The material shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules and Container Specifications for rail shipments or National Motor Freight Rules and Container Specifications for truck shipment as applicable.

5.2 Marking.-In addition to any special markings required by the contract or purchase order, containers shall be marked in accordance with MIL-STD-129, and ICC regulations "United P" and 18-55-56. Markings shall include, but not be limited to, the following information:

a. Manufacturer's name
b. Product designation
c. Lot number
d. Date of manufacture
e. Number of this specification

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this document.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Intended use.-The chloro-fluoro hydrocarbons covered in this specification are intended to be used in ammunition items.

6.4 The material described in this specification can be obtained from Allied Chemical Corp., Dupont Corp., Pennsalt Chemicals Corp., and Union Carbide Corp. Materials from other sources must be qualified for use in the intended end item.

6.5 Solvents recovered during the process and re-used may contain a maximum of 100 ppm of water.

Custodian:
Army-MU

Preparing activity:
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